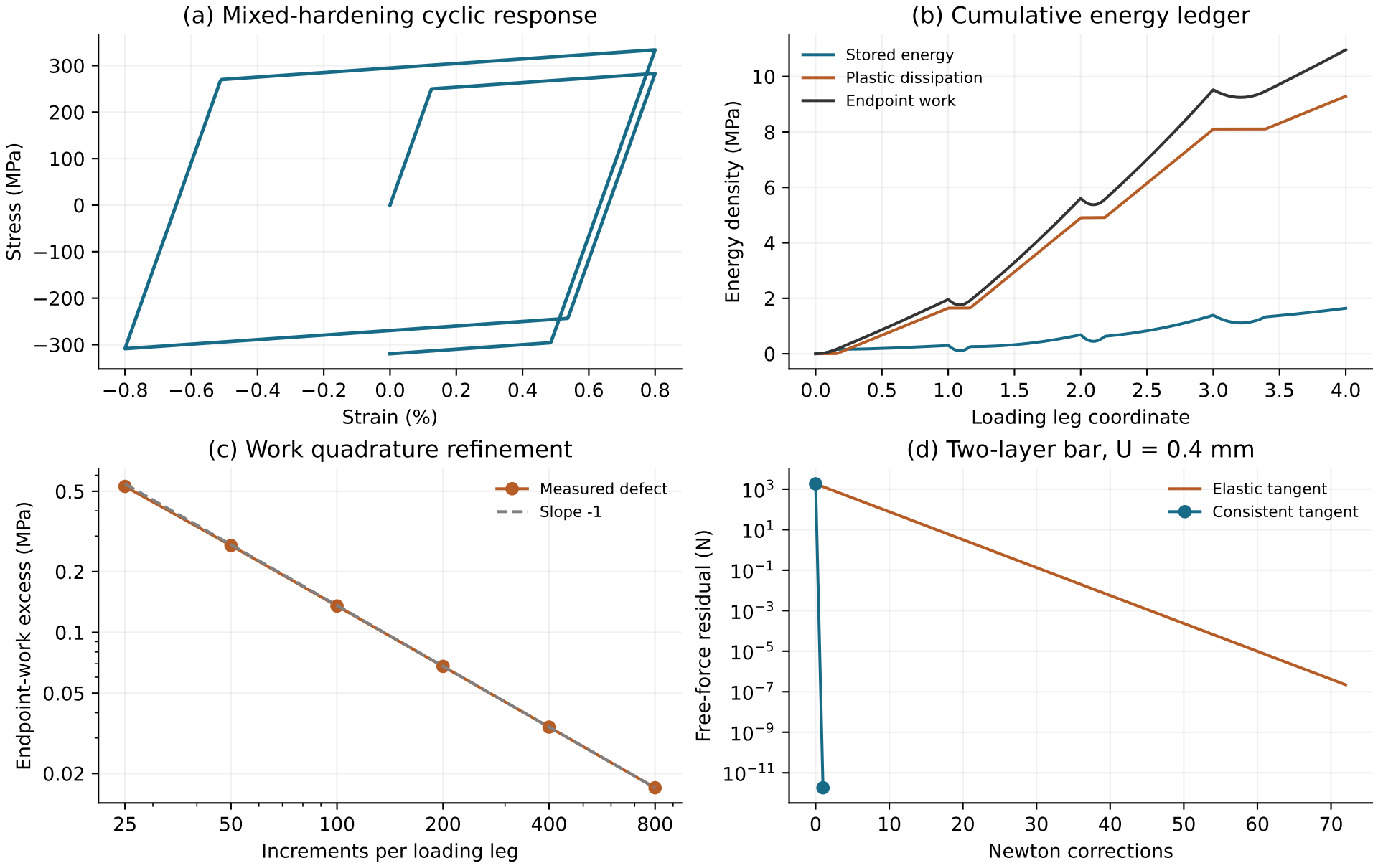
(a) Mixed-hardening cyclic response
Stress (MPa)
Strain (%)
(b) Cumulative energy ledger
Stored energy
Plastic dissipation
Endpoint work
Energy density (MPa)
Loading leg coordinate
(c) Work quadrature refinement
Measured defect
Slope -1
Endpoint-work excess (MPa)
Increments per loading leg
(d) Two-layer bar, U = 0.4 mm
Elastic tangent
Consistent tangent
Free-force residual (N)
Newton corrections